CI/CD Setup Guide: GitHub → Azure Container Registry

Beginner - Friendly Guide to CI/CD from GitHub to Azure Container Registry using OIDC

This guide walks you through setting up GitHub Actions to automatically build and push your Docker image to Azure Container Registry (ACR). It uses secure OIDC-based authentication with no stored secrets. Every push to your repository will trigger this workflow.

What You'll End Up With

Each push to your branch (e.g., dev) builds and pushes the Docker image to ACR, with tags like:

<your-registry>.azurecr.io/<PROJECT-NAME>:latest

<your-registry>.azurecr.io/<PROJECT-NAME>:sha-ABC1234

This is done using OIDC (OpenID Connect) without storing any client secrets.

One-Time Setup in Azure (10–15 mins)

1) Create an App Registration for GitHub

Steps via Azure Portal:

• Go to Microsoft Entra ID → App registrations → New registration.

• Name: github-oidc-<PROJECT-NAME>

• Supported account types: Single tenant → Register

Copy the following values:

• Application (client) ID

• Directory (tenant) ID

• Subscription ID

Now add a Federated Credential:

• Go to Certificates & secrets → Federated credentials → Add credential.

• Issuer: GitHub

• Organization: <your-org>

• Repository: <PROJECT-NAME>

• Entity Type: Branch

• Based on Selection: dev

2) Assign AcrPush Role to the App

• Open your Container Registry → Access control (IAM) → Add role assignment.

• Role: AcrPush

• Assign access to: User, group, or service principal

• Pick the app you registered → Save

One-Time Setup in GitHub (5 mins)

Go to your repo → Settings → Secrets and variables → Actions.

Add the following Secrets:

• AZURE_TENANT_ID

• AZURE_SUBSCRIPTION_ID

• AZURE_CLIENT_ID

Add the following Variables:

• ACR_NAME → your registry name (e.g., myregistry)

• ACR_LOGIN_SERVER → your login server (e.g., myregistry.azurecr.io)

CI Workflow File: .github/workflows/build-and-push.yml

Paste the following YAML into a new file: Download YML file

Verify it Worked

• In GitHub → Actions → latest run → verify build and push steps

• In Azure Portal → ACR → Repositories → <PROJECT-NAME>

Local Sanity Check (Optional)

Run your app locally via Docker:

```bash
docker build -t test-<PROJECT-NAME> .
docker run -p 8000:8000 test-<PROJECT-NAME>
```

Troubleshooting Tips

• Check AcrPush is correctly assigned

• Ensure `id-token: write` is in permissions

• Federated credential must match branch name

• Image tag must not be blank